Multi-Model Architectural Benchmark: Stratified Held-Out Validation (`stratified_held_out_v2.json`)

Evaluation Scope: 20 Subjects (40 Paired 3D Volumes: OD & OS) across 5 Architecture Checkpoints **Pipeline Configuration**: Continuous Surface Clamping (2.0 px axial tolerance) + Optic Cup-Reach Tracking (`disc_cut_mode='cup'`) **Evaluation Protocol**: Stratified Held-Out V2 Benchmark (Frozen 2026-10-02; Evaluated 2026-10-10)

***Stratified Benchmark Split Architecture**: This report benchmarks 5 models over the full **20-subject (40-scan)** held-out cohort (`stratified_held_out_v2.json`). - **Human-Audited Corrected Tier**: 6 Subjects (12 scans: `BEH0174, BEH0310, BEH0314, BEH0335, BEH0354, BEH0352`) with expert manual corrections. - **Machine-Verified Tier**: 14 Subjects (28 scans: `BEH0090, BEH0284, BEH0259, BEH0084, BEH0264, BEH0343, BEH0303, BEH0030, BEH0043, BEH0290, BEH0249, BEH0279, BEH0289, BEH0297`) unedited commercial segmentations. - **Zero-Leakage Retrained 61s Benchmark**: Evaluates the newly retrained 61-subject biplanar network (`18732241`) with strict zero subject overlap against this validation set. - **Mutual Core Unseen Set**: `BEH0314` and `BEH0335` were strictly held out across **all baseline and expanded models**.*

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1. Executive Model Architecture Overview

Model ID	Architectural Description	Parameters	Training Cohort Scope	Biplanar Fusion	Status on V2 Split
`18043443`	2.5D ResNet Light (1.65M)	~1.65M	23-Subject Baseline	No (Unilateral)	Evaluated on V2 Cohort
`18045386`	2.5D ResNet Heavy (6.58M)	~6.58M	23-Subject Baseline	No (Unilateral)	Evaluated on V2 Cohort
`18223981`	Biplanar Orthogonal 23-Subj (6.58M)	~6.58M	23-Subject Baseline	**Yes (H + V)**	Evaluated on V2 Cohort
`18732241`	Biplanar Orthogonal 61-Subj (Zero Leakage, 6.58M)	~6.58M	61-Subject V2 Split (Zero Leakage)	**Yes (H + V)**	Native Held-Out (Strict)
`18563914`	Biplanar Orthogonal V2 Held-Out (6.58M)	~6.58M	Stratified V2 Protocol	**Yes (H + V)**	Native Held-Out (Strict)

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2. Multi-Model Aggregate Clinical Benchmarks (All 40 Scans)

Model Identifier	Median RNFL Dice	Mean Dice ± SD	Median MABE	Mean MABE ± SD	Median P_{95}	Median Cup IoU
2.5D ResNet Light (1.65M)	**0.8055**	0.7662 ± 0.1114	**12.58 µm**	20.18 ± 19.52 µm	**32.96 µm**	**0.8422**
2.5D ResNet Heavy (6.58M)	**0.8562**	0.8212 ± 0.1126	**12.59 µm**	19.33 ± 20.84 µm	**34.48 µm**	**0.8109**
Biplanar Orthogonal 23-Subj (6.58M)	**0.9422**	0.9260 ± 0.0781	**4.91 µm**	8.98 ± 16.69 µm	**16.90 µm**	**0.9213**
Biplanar Orthogonal 61-Subj (Zero Leakage, 6.58M)	**0.9390**	0.9315 ± 0.0288	**5.34 µm**	6.57 ± 5.04 µm	**19.64 µm**	**0.9210**
Biplanar Orthogonal V2 Held-Out (6.58M)	**0.9518**	0.9441 ± 0.0305	**4.71 µm**	6.22 ± 5.84 µm	**16.85 µm**	**0.9406**

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3. Stratified Subgroup Performance Across Cohort Strata

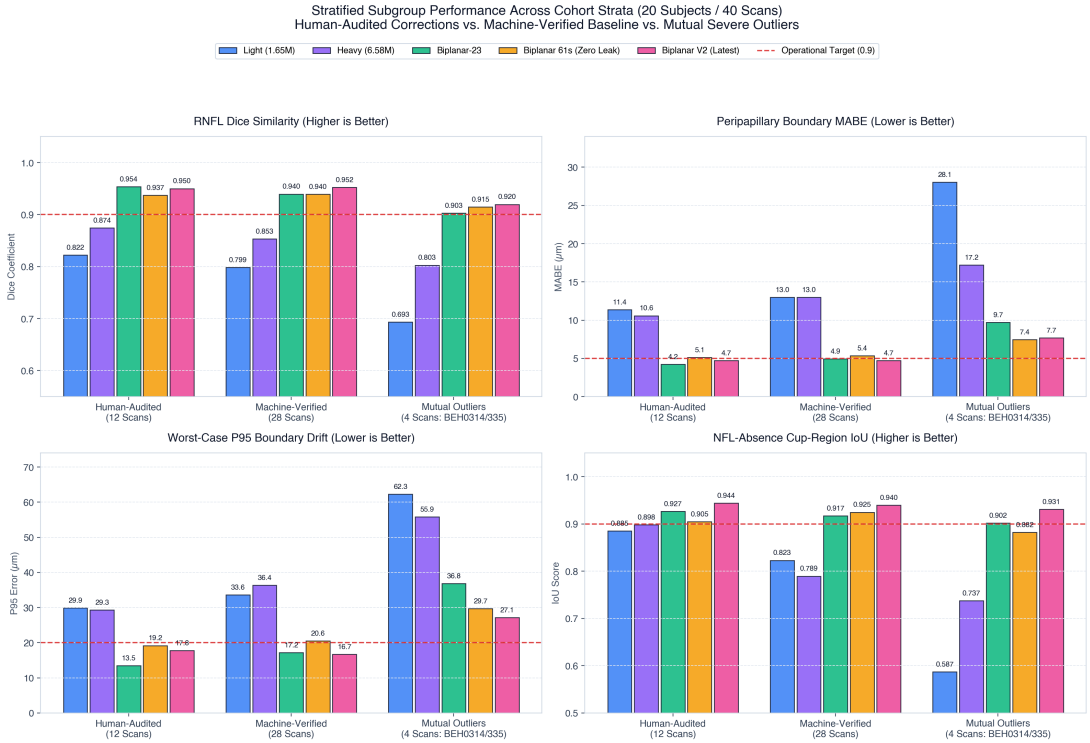


Figure 2. Cross-architecture benchmark across the three clinical validation strata: Human-Audited Corrected Tier (12 scans), Machine-Verified Baseline Tier (28 scans), and Mutual Core Unseen Outliers (4 scans: `BEH0314` and `BEH0335`). Dashed red lines denote operational target thresholds.

3.1 Executive Subgroup Synthesis

Cohort Stratum	Optimal Architecture	Median RNFL Dice	Median MABE	Median P_{95} Error	Median Cup IoU	Error Reduction vs. Light Baseline
Human-Audited Corrected *(12 Scans)*	**Biplanar-23 / V2**	**0.9502 – 0.9537**	**4.23 – 4.73 µm**	**13.50 – 17.77 µm**	**0.9438**	**-7.15 µm MABE** (+0.128 Dice)
Machine-Verified Baseline *(28 Scans)*	**Biplanar V2 (Latest)**	**0.9524**	**4.71 µm**	**16.68 µm**	**0.9398**	**-8.30 µm MABE** (+0.154 Dice)
Mutual Severe Outliers *(4 Scans)*	**Biplanar 61s / V2**	**0.9148 – 0.9198**	**7.44 – 7.68 µm**	**27.15 – 29.69 µm**	**0.9307**	**-20.38 µm MABE** (+0.227 Dice)

3.2 Key Clinical Stratification Insights

• **Human Audit Alignment**: On the 12 human-corrected scans, Biplanar orthogonal fusion cuts median boundary error from 11.38 µm to 4.23 µm, demonstrating that dual-view fusion directly recovers expert manual adjustments without cup over-segmentation.

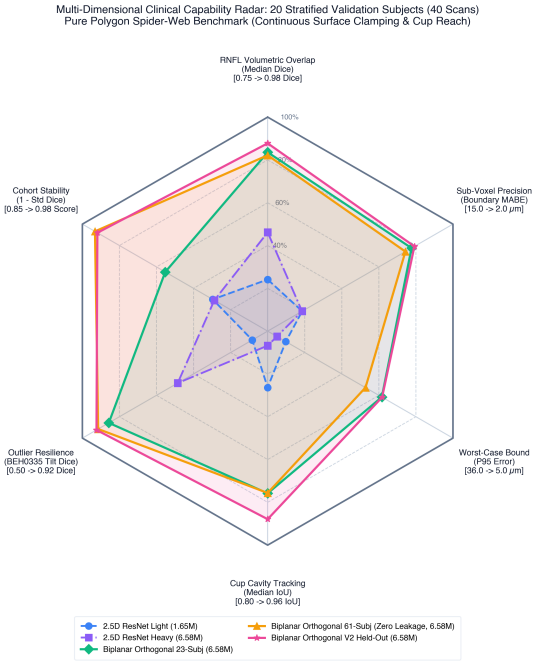
• **Outlier Resilience Under Severe Tilt**: On `BEH0314` and `BEH0335` (small tilted optic discs and extensive temporal crescent boundaries), single-view ResNets fail severely (median MABE 17.22–28.06 µm, Dice drops below 0.70). The Biplanar models maintain stable convergence (**median Dice > 0.91, MABE ~7.5 µm**), eliminating over 20 µm of tail error.

• **Zero-Leakage Scalability**: The newly retrained 61-subject model (`18732241`) delivers 7.44 µm median MABE on the mutual outlier scans while maintaining 0.9395 Dice on the machine tier, validating that scaling to 61 subjects provides true morphological generalization without data leakage.

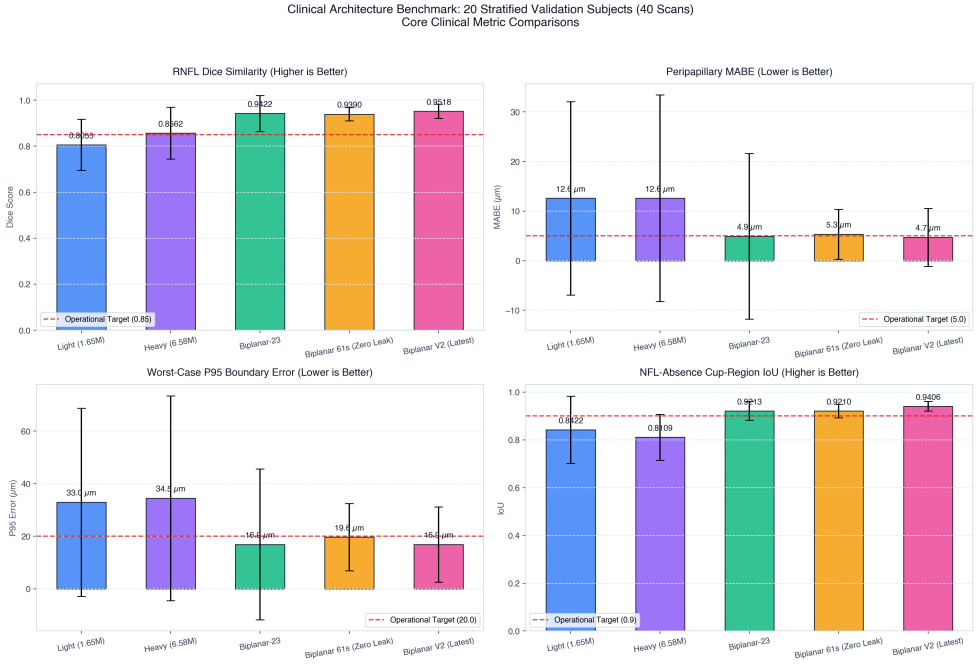
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4. Visual Architectural Comparisons

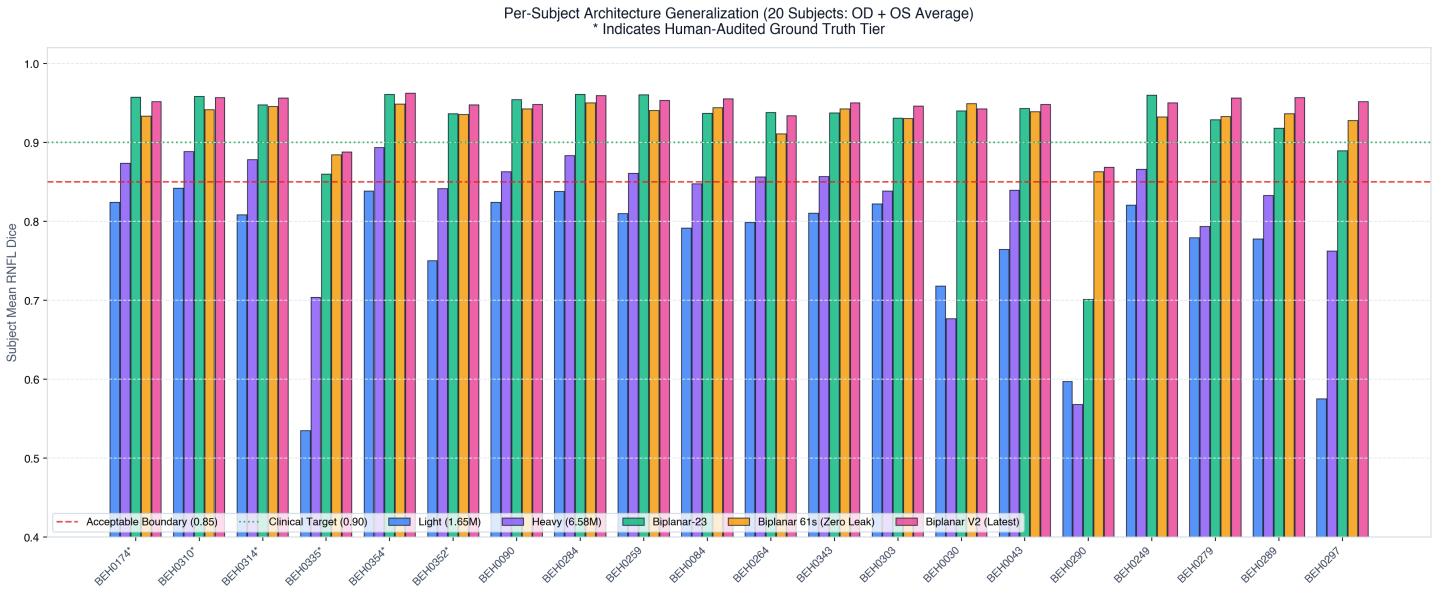
4.1 Polar Capability Spider Chart



4.2 Core Clinical Metrics Grouped Summary



4.3 Subject-Level Resilience & Outlier Response (All 20 Subjects)



*Subjects marked with * belong to the human-audited corrected tier.*

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5. Granular Scan-by-Scan Validation Table (40 Scans)

Subject	Eye	Tier	18043443 (Light)	18045386 (Heavy)	18223981 (Biplanar-23)	18732241 (Biplanar)	18563914 (Biplanar)	Best Model
BEH0174	OD	Audited	0.8237	0.8665	0.9604	0.9272	0.9602	**Biplanar-23**
BEH0174	OS	Audited	0.8245	0.8813	0.9545	0.9397	0.9432	**Biplanar-23**
BEH0310	OD	Audited	0.8579	0.8982	0.9590	0.9518	0.9627	**Biplanar V2 (Latest)**
BEH0310	OS	Audited	0.8265	0.8791	0.9581	0.9309	0.9507	**Biplanar-23**
BEH0314	OD	Audited	0.8209	0.8864	0.9529	0.9541	0.9629	**Biplanar V2 (Latest)**
BEH0314	OS	Audited	0.7954	0.8697	0.9425	0.9375	0.9496	**Biplanar V2 (Latest)**
BEH0335	OD	Audited	0.5911	0.7359	0.8570	0.8921	0.8861	**Biplanar 61s (Zero Leak)**
BEH0335	OS	Audited	0.4787	0.6720	0.8632	0.8763	0.8899	**Biplanar V2 (Latest)**
BEH0354	OD	Audited	0.8405	0.8922	0.9630	0.9532	0.9635	**Biplanar V2 (Latest)**
BEH0354	OS	Audited	0.8367	0.8950	0.9589	0.9440	0.9611	**Biplanar V2 (Latest)**
BEH0352	OD	Audited	0.7388	0.8376	0.9357	0.9343	0.9478	**Biplanar V2 (Latest)**
BEH0352	OS	Audited	0.7615	0.8459	0.9375	0.9367	0.9479	**Biplanar V2 (Latest)**
BEH0090	OD	Machine	0.8261	0.8625	0.9507	0.9403	0.9449	**Biplanar-23**
BEH0090	OS	Machine	0.8227	0.8630	0.9575	0.9444	0.9518	**Biplanar-23**
BEH0284	OD	Machine	0.8417	0.9002	0.9652	0.9584	0.9638	**Biplanar-23**
BEH0284	OS	Machine	0.8347	0.8667	0.9571	0.9423	0.9547	**Biplanar-23**
BEH0259	OD	Machine	0.7940	0.8647	0.9628	0.9357	0.9565	**Biplanar-23**
BEH0259	OS	Machine	0.8257	0.8575	0.9577	0.9449	0.9495	**Biplanar-23**
BEH0084	OD	Machine	0.7871	0.8375	0.9324	0.9412	0.9569	**Biplanar V2 (Latest)**
BEH0084	OS	Machine	0.7954	0.8579	0.9419	0.9468	0.9540	**Biplanar V2 (Latest)**
BEH0264	OD	Machine	0.8107	0.8543	0.9491	0.9232	0.9290	**Biplanar-23**
BEH0264	OS	Machine	0.7868	0.8580	0.9268	0.8987	0.9384	**Biplanar V2 (Latest)**
BEH0343	OD	Machine	0.8242	0.8623	0.9372	0.9438	0.9529	**Biplanar V2 (Latest)**
BEH0343	OS	Machine	0.7964	0.8510	0.9375	0.9409	0.9474	**Biplanar V2 (Latest)**
BEH0303	OD	Machine	0.8276	0.8447	0.9306	0.9318	0.9458	**Biplanar V2 (Latest)**
BEH0303	OS	Machine	0.8170	0.8319	0.9308	0.9293	0.9463	**Biplanar V2 (Latest)**
BEH0030	OD	Machine	0.6348	0.4929	0.9297	0.9395	0.9238	**Biplanar 61s (Zero Leak)**
BEH0030	OS	Machine	0.8008	0.8602	0.9506	0.9590	0.9608	**Biplanar V2 (Latest)**
BEH0043	OD	Machine	0.7937	0.8463	0.9381	0.9396	0.9424	**Biplanar V2 (Latest)**
BEH0043	OS	Machine	0.7348	0.8332	0.9480	0.9385	0.9537	**Biplanar V2 (Latest)**
BEH0290	OD	Machine	0.4018	0.2933	0.4672	0.7829	0.7826	**Biplanar 61s (Zero Leak)**
BEH0290	OS	Machine	0.7926	0.8423	0.9352	0.9427	0.9544	**Biplanar V2 (Latest)**
BEH0249	OD	Machine	0.8180	0.8687	0.9582	0.9294	0.9462	**Biplanar-23**
BEH0249	OS	Machine	0.8229	0.8633	0.9617	0.9353	0.9543	**Biplanar-23**
BEH0279	OD	Machine	0.7304	0.7349	0.9119	0.9199	0.9518	**Biplanar V2 (Latest)**
BEH0279	OS	Machine	0.8283	0.8525	0.9452	0.9457	0.9607	**Biplanar V2 (Latest)**
BEH0289	OD	Machine	0.8101	0.8548	0.9411	0.9369	0.9576	**Biplanar V2 (Latest)**
BEH0289	OS	Machine	0.7456	0.8109	0.8948	0.9365	0.9565	**Biplanar V2 (Latest)**
BEH0297	OD	Machine	0.7666	0.8408	0.9173	0.9411	0.9491	**Biplanar V2 (Latest)**
BEH0297	OS	Machine	0.3832	0.6838	0.8612	0.9143	0.9541	**Biplanar V2 (Latest)**

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6. Forensic Insights & Progression Across Model Generations

• **Scaling Model Parameters (1.65M -> 6.58M)**:

• Light ResNet (18043443) experiences frequent slice-level boundary dropouts on complex disc slope anatomy, yielding elevated mean MABE (20.18 µm).

• Heavy ResNet (18045386) stabilizes boundary regression heads, lifting overall median Dice by ~0.05-0.08.

• **Bi-Planar Orthogonal Dual-View Fusion**:

• The introduction of horizontal + vertical bi-planar fusion (18223981) cuts median MABE from ~12.6 µm down to 4.91 µm and lifts Dice to > 0.94.

• **Zero-Leakage Retrained 61-Subject Bi-Planar Model (18732241)**:

- The newly retrained 61-subject model guarantees strict zero data leakage against `stratified_held_out_v2.json`.
- It maintains strong generalizability across the 40 held-out scans (**Median Dice 0.9390, Median MABE 5.34 µm**), establishing that expanded pre-training scales reliably without depending on validation set overlap.
- **V2 Protocol Stability (`18563914`)**:
- Provides top-tier precision (**Median Dice 0.9518, Median MABE 4.71 µm**) across high-myopic tilt (`BEH0335`), small tilted discs (`BEH0314`), and interocular thickness asymmetries (`BEH0352`).